Tuesday, October 27, 2026

1 Heteroskedasticity

$$\underset{\sim}{\varepsilon} \sim N(\underset{\sim}{0}, \Sigma)$$
$$\overset{n\times n}{\Sigma} \neq \sigma^2 I$$

If we can estimate Σ satisfactorily by $\hat{\Sigma}$, then we can apply GLS.

$$\underset{\sim}{Y} \longrightarrow \hat{\Sigma}^{-1/2}\underset{\sim}{Y}$$
$$X \longrightarrow \hat{\Sigma}^{-1/2}X$$

But, in order to estimate Σ, (without any structure) we need to estimate $n(n+1)/2$ components, which is NOT possible with n independent samples.

Therefore, we assume some structure on Σ:

Assumed Structures on Σ

I) $\Sigma = \text{Diag}\left(\sigma_1^2, \ldots, \sigma_n^2\right)$

(a) $\sigma_i^2 = w\left(\underset{\sim i}{z}, \overset{k\times 1}{\underset{\sim}{\lambda}}\right)$ where w is known and $\underset{\sim i}{z}$'s are observed.

(b) $\sigma_i^2 = g\left(\underset{\sim i}{x}^T \underset{\sim}{\beta}\right)$ where g is known.

II) Variance is same for all ε_i, but they are correlated.

– Further assumptions are required.

1.1 Case I(a): $\sigma_i^2 = w(\underset{\sim i}{z}, \underset{\sim}{\lambda})$

I) a)

$$l_n\left(\underset{\sim}{\beta}, \underset{\sim}{\lambda}\right) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{n}\log\sigma_i^2 - \frac{1}{2}\sum_{i=1}^{n}\frac{\left(y_i - \underset{\sim i}{x}^T\underset{\sim}{\beta}\right)^2}{\sigma_i^2} \quad ; \qquad \sigma_i^2 = w\left(\underset{\sim i}{z}, \underset{\sim}{\lambda}\right).$$

$$\frac{\partial l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)}{\partial \underset{\sim}{\beta}} = \underset{\sim}{0} \implies \underset{\sim}{\hat{\beta}} = \left(X^T\hat{\Sigma}^{-1}X\right)^{-1}X^T\hat{\Sigma}^{-1}\underset{\sim}{Y} \tag{1}$$

$$\begin{aligned}\frac{\partial l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)}{\partial \underset{\sim}{\lambda}} &= \sum_{i=1}^{n}\frac{\partial l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)}{\partial\sigma_i^2}\cdot\frac{\partial\sigma_i^2}{\partial\underset{\sim}{\lambda}}\\ &= \sum_{i=1}^{n}\left\{-\frac{1}{2\sigma_i^2}+\frac{1}{2\sigma_i^4}\underbrace{\left(y_i-\underset{\sim i}{x}^T\underset{\sim}{\hat{\beta}}\right)^2}_{e_i^2}\right\}\frac{\partial\sigma_i^2}{\partial\underset{\sim}{\lambda}} = \underset{\sim}{0}.\end{aligned} \tag{2}$$

Usually, ① and ② are not solvable analytically, one has to employ an iterative algorithm:

- Fixing $\lambda = \hat{\lambda}$, define $\hat{\Sigma} = \Sigma(\underset{\sim}{\hat{\lambda}})$ and update $\beta = \hat{\beta} = \left(X^T\hat{\Sigma}^{-1}X\right)^{-1}X^T\hat{\Sigma}^{-1}\underset{\sim}{Y}$
- Fixing $\beta = \hat{\beta}$, update λ.

Profile Likelihood

$$l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)$$

$$\underline{\underline{l_p\left(\underset{\sim}{\beta}\right)}} = \max_{\underset{\sim}{\lambda}}\ \underline{l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)}$$

$$\max_{\underset{\sim}{\beta}}\boxed{l_p\left(\underset{\sim}{\beta}\right)} = \underline{\underline{\max_{\underset{\sim}{\beta}}\ \max_{\underset{\sim}{\lambda}}\ l_n\left(\underset{\sim}{\beta},\underset{\sim}{\lambda}\right)}}.$$

Example: $\sigma_i^2 = \exp\left\{\underset{\sim i}{z}^T\underset{\sim}{\lambda}\right\}$ $\qquad$ $w\left(\underset{\sim i}{z},\underset{\sim}{\lambda}\right) = \exp\left\{\underset{\sim i}{z}^T\underset{\sim}{\lambda}\right\}$.

Fix $\underset{\sim}{\beta}$, $\left(\text{Note: } \dfrac{\partial \sigma_i^2}{\partial \underset{\sim}{\lambda}} = \sigma_i^2 \cdot \underset{\sim}{z}_i\right)$

$$\frac{\partial l_n\left(\underset{\sim}{\lambda}\right)}{\partial \underset{\sim}{\lambda}} = -\sum_{i=1}^{n} \frac{1}{\cancel{\sigma_i^2}} \left\{1 - \frac{e_i^2(\underset{\sim}{\beta})}{\sigma_i^2(\underset{\sim}{\lambda})}\right\} \cancel{\sigma_i^2} \underset{\sim}{z}_i \tag{3}$$

$$\frac{\partial^2 l_n\left(\underset{\sim}{\lambda}\right)}{\partial \underset{\sim}{\lambda} \partial \underset{\sim}{\lambda}'} = \sum_{i=1}^{n} \frac{\partial}{\partial \sigma_i^2} \left(\frac{\partial l_n\left(\underset{\sim}{\lambda}\right)}{\partial \underset{\sim}{\lambda}}\right) \cdot \underbrace{\frac{\partial \sigma_i^2}{\partial \underset{\sim}{\lambda}}}_{\sigma_i^2 \underset{\sim}{z}_i}$$

$$= -\sum_{i=1}^{n} \frac{e_i^2(\underset{\sim}{\beta})}{\cancel{\sigma_i^4}\cancel{\sigma_i^2}} \underset{\sim}{z}_i \underset{\sim}{z}_i^T \cancel{\sigma_i^2}$$

$$= -\underbrace{Z^T \text{Diag}\left(\frac{e_i^2(\underset{\sim}{\beta})}{\sigma_i^2}, i = 1, \ldots, n\right)^{n\times k} Z}_{\text{[negative definite matrix]}} \qquad Z = \begin{bmatrix} \underset{\sim}{z}_1' \\ \vdots \\ \underset{\sim}{z}_n' \end{bmatrix}$$

The soln. of ③ is indeed a maxima.

Testing

$$H_0 : \sigma_1^2 = \cdots = \sigma_n^2 = \sigma^2.$$

- **Under H_0:** $l_{n,0}^*\left(\underset{\sim}{\beta}, \sigma^2\right) = -\dfrac{n}{2}\log(2\pi) - \dfrac{n}{2}\log\hat{\sigma}_{ML}^2 \qquad \hat{\sigma}_{ML}^2 = \dfrac{RSS}{n}$
- **Under H_1 / without any restriction:** $l_n^*\left(\underset{\sim}{\beta}, \underset{\sim}{\lambda}\right) = l_n\left(\hat{\underset{\sim}{\beta}}_{opt}, \hat{\underset{\sim}{\lambda}}_{opt}\right)$

  where $\left(\hat{\underset{\sim}{\beta}}_{opt}, \hat{\underset{\sim}{\lambda}}_{opt}\right)$ is the soln. to the iterative algorithm.

$T(X) = -2l_{n,0}^*\left(\underset{\sim}{\beta}, \sigma^2\right) + 2l_n^*\left(\underset{\sim}{\beta}, \underset{\sim}{\lambda}\right) \xrightarrow[H_0]{d} \chi^2_{\underline{k-1}}$ as $n \to \infty$. $\qquad (\cancel{p} + k + 1) - (\cancel{p} + 2) = k - 1$

Breusch-Pagan Test

When the form of w is not known, then one can employ the Breusch-Pagan test.

Bruesch-Pagan test: $\left[\text{Let } \sigma_i^2 = \underset{\uparrow}{w}_{\text{unknown}}\left(\lambda_0 + \underset{\sim}{z}_i^T \underset{\sim}{\lambda}\right) ;\ H_0 : \underline{\underset{\sim}{\lambda} = \underset{\sim}{0}}\right]$

- First fit OLS with the data, and find $e_i,\ i = 1, \ldots, n$.
- Define $u_i = e_i^2/\hat{\sigma}_{ML}^2$.
- Regress u_i w.r.t. $\underset{\sim}{z}_i$ (OLS) and find the fit $\hat{\underset{\sim}{u}} = P(Z)\underset{\sim}{u}$.

– Test statistic: $S_0^2 = \frac{1}{2}\sum_{i=1}^{n}\left(\hat{u}_i - \underbrace{\bar{u}}_{\hat{\bar{u}}}\right)^2$.

– Under H_0: $S_0^2 \xrightarrow{d} \chi_k^2$.

– Reject for higher values of S_0^2.

1.2 Case I(b): $\sigma_i^2 = g(\underset{\sim i}{x}^T \underset{\sim}{\beta})$

I) b) $\sigma_i^2 = g\left(\underset{\sim i}{x}^T \underset{\sim}{\beta}\right)$ and g is known.

Algorithm:

1) Fix $\underset{\sim}{\beta} = \underset{\sim(j)}{\beta}$, and define $\hat{\Sigma}_{(j)} = \text{Diag}\left(g\left(\underset{\sim i}{x}^T \underset{\sim(j)}{\beta}\right),\ j = 1, \ldots, n\right)$ at the jth iteration.

2) $\underset{\sim(j+1)}{\hat{\beta}} = \left(X^T\hat{\Sigma}_{(j)}^{-1}X\right)^{-1} X^T\hat{\Sigma}_{(j)}^{-1}X$; then set $j = j + 1$.

It has been established that the soln. of $\underset{\sim}{\beta}$ obtained from the above algorithm has the same asymptotic efficiency as $\underset{\sim WLS}{\hat{\beta}}$ when the Σ were known.

Thus, there is asymptotically no loss for not knowing Σ exactly.

Cook-Weisberg Method

When the form of g is not known, then one may apply Cook-Weisberg's method.
Cook-Weisberg: Similar to Bruesh-Pagan test.

– Define u_i as before,

– $\left(\underline{\underline{\text{OLS}}}\right)$ regress u_i w.r.t. $\underline{\underline{\hat{y}_i}}$.

– Define S_0^2 similarly as before.

– Under H_0, $S_0^2 \xrightarrow{d} \chi_1^2$, as $n \to \infty$.

2 Correlated Errors, with Same Variance

II) Correlated errors, with same variance.

As there are $\frac{n(n-1)}{2} + 1$ components to estimate in Σ, which is not possible, we need to consider further assumptions.

Index: $t = 1, 2, \ldots, T$.

Stationarity

Stationarity:

$$\begin{cases} E(\varepsilon_t) = 0 \quad \forall\ t = 1, \ldots, T \\ \text{Var}(\varepsilon_t) = \sigma_\varepsilon^2 \quad \forall\ t = 1, \ldots, T. \\ \underline{\text{Corr}}\,(\varepsilon_t, \varepsilon_{t+h}) = \underline{\underline{\rho_h}} \quad \forall\ t \text{ and } h. \end{cases}$$

$$\Sigma = \sigma_\varepsilon^2 \begin{bmatrix} 1 & \rho_1 & \rho_2 & \cdots \\ \rho_1 & \underline{1} & \rho_1 & \cdots \\ \rho_2 & \rho_1 & 1 & \\ \vdots & & & \ddots \end{bmatrix} \qquad \text{there are } T \text{ unknowns.}$$

So, we have to take more assumptions.

2.1 AR(1) Error Structure

$$\underline{\varepsilon_t = \rho\varepsilon_{t-1} + u_t} \quad ; \quad t = 2, \ldots, T. \qquad \begin{bmatrix} u_2 \\ \vdots \\ u_T \end{bmatrix} \sim N\left(\underset{\sim}{0}, \sigma^2 I\right)$$

$$\varepsilon_0 = 0$$

$$\varepsilon_1 = \rho \times 0 + u_1 = u_1. \qquad u_1 \perp\!\!\!\perp \begin{pmatrix} u_2 \\ \vdots \\ u_T \end{pmatrix} \qquad u_1 \sim N\left(0, \frac{\sigma^2}{1-\rho^2}\right).$$

$$\begin{aligned} \text{Var}(\varepsilon_1) &= \text{Var}(u_1) = \sigma_\varepsilon^2 \\ \text{Var}(\varepsilon_2) &= \text{Var}\left(\rho u_1 + u_2\right) = \rho^2 \text{Var}(u_1) + \text{Var}(u_2) + 2\rho\underline{\text{Cov}(u_1, u_2)} \\ \sigma_\varepsilon^2 &= \rho^2\sigma_\varepsilon^2 + \sigma^2 \\ \implies \sigma_\varepsilon^2\left(1-\rho^2\right) &= \sigma^2 \\ \implies \sigma_\varepsilon^2 &= \frac{\sigma^2}{1-\rho^2} \\ \text{Var}(\varepsilon_t) &= \sigma_\varepsilon^2. \\ \underline{\underline{\text{Cov}\,(\varepsilon_t, \varepsilon_{t+h})}} &= ? \end{aligned}$$

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$\underline{\varepsilon_t = \rho\varepsilon_{t-1} + u_t \ ; \ t = 2, \ldots, T.}$ $\quad \begin{bmatrix} u_2 \\ \vdots \\ u_T \end{bmatrix} \sim N\left(\underset{\sim}{0}, \sigma^2 I\right)$

$\text{Var}(\varepsilon_t) = \sigma^2_\varepsilon = \sigma^2/(1-\rho^2) \qquad u_1 \perp\!\!\!\perp \begin{pmatrix} u_2 \\ \vdots \\ u_T \end{pmatrix}$

$\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = \rho^h \dfrac{\sigma^2}{1-\rho^2} \qquad u_1 \sim N\left(0, \dfrac{\sigma^2}{1-\rho^2}\right).$

Stationary —
$E(\varepsilon_t) = 0 \qquad \forall\ t$
$\text{Var}(\varepsilon_t)$ is same for all t.
$\underline{\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = \rho(h)}$

Derivations:

$$\begin{aligned} \text{Var}(\varepsilon_t) &= \text{Var}\left(\rho\varepsilon_{t-1} + u_t\right) \\ &= \rho^2 \underline{\text{Var}(\varepsilon_{t-1})} + \sigma^2 + 2\rho \underbrace{\text{Cov}\left(\varepsilon_{t-1}, u_t\right)}_{0 \quad -(*)} \end{aligned}$$

where for $(*)$:

$$\begin{aligned} \text{Cov}\left(\varepsilon_{t-1}, u_t\right) &= \text{Cov}\left(\rho\varepsilon_{t-2} + u_{t-1}, u_t\right) \\ &= \rho\text{Cov}\left(\varepsilon_{t-2}, u_t\right) = \cdots = \rho^{t-1}\text{Cov}\Big(\underset{\underset{u_1}{\|}}{\varepsilon_1}, u_t\Big) = \rho^{t-1} \times 0 = 0 \end{aligned}$$

Thus:

$$\hookrightarrow \text{Var}(\varepsilon_t) = \rho^2 \frac{\sigma^2}{1-\rho^2} + \sigma^2 = \frac{\sigma^2}{1-\rho^2}.$$

And for the covariance:

$$\text{Cov}\left(\varepsilon_t, \varepsilon_{t+h}\right) = \text{Cov}\left(\varepsilon_t, \rho\varepsilon_{t+h-1} + u_{t+h}\right) = \rho\text{Cov}\left(\varepsilon_t, \varepsilon_{t+h-1}\right) = \cdots = \rho^h \text{Var}(\varepsilon_t) = \rho^h \frac{\sigma^2}{1-\rho^2}$$

$$\Sigma_T = \frac{\sigma^2}{1-\rho^2} \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{T-1} \\ \rho & 1 & \rho & \cdots & \rho^{T-2} \\ \vdots & & & & \vdots \\ \rho^{T-1} & \rho^{T-2} & \cdots & & 1 \end{bmatrix} = \Sigma\left(\sigma^2, \rho\right).$$

2.2 Diagnostic Plots for Correlated Errors

Plots

I) Plot of the error. $\{e_t \ ; \ t = 1, \ldots, T\}$.

$$\left[\text{Note: } \underset{\sim}{e} \sim N\left(\underset{\sim}{0}, I - P(X)\right), \ I - P(X) \approx I, \ p_{ii} \sim p/n, \ p_{ij} \sim -1/n\right]$$

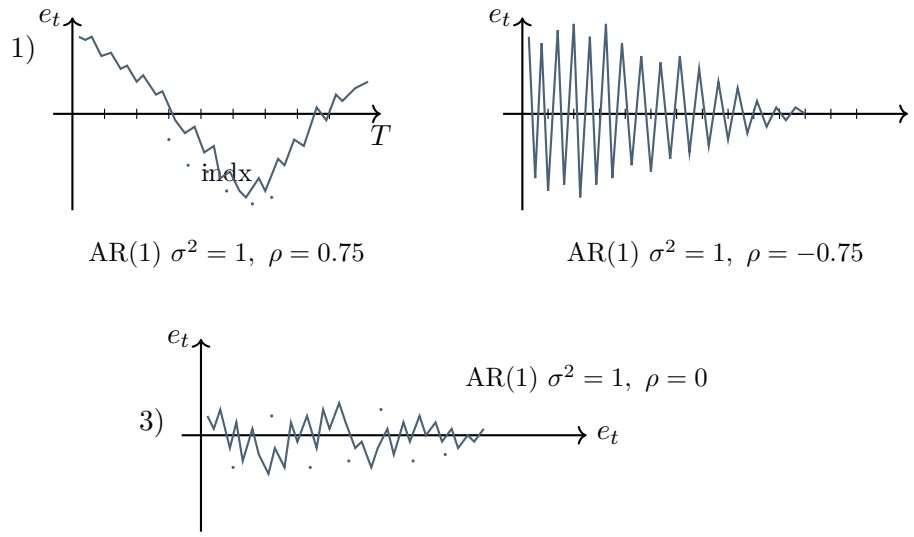


Figure 1: Time series plots of residuals $\{e_t\}$ under AR(1) for $\rho = 0.75$, $\rho = -0.75$, and $\rho = 0$.

II ACF: Autocorrelation-function Plot.

$$\text{Cov}\,(\varepsilon_t, \varepsilon_{t-h}) \qquad h = 1, 2, \ldots$$

$$\hat{\rho}_h = \frac{\sum_{t=h+1}^{T} e_t e_{t-h}}{\sum_{t=1}^{T} e_t^2} = \frac{\sum_{t=h+1}^{T} (e_t - \bar{e})(e_{t-h} - \bar{e})}{\sum_{t=1}^{T} (e_t - \bar{e})^2}.$$

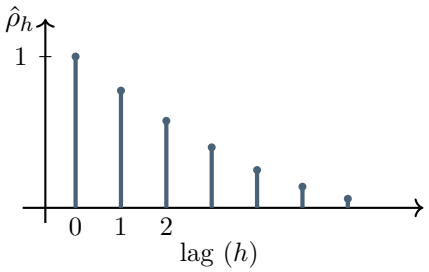


Figure 2: ACF Plot showing geometric decay of $\hat{\rho}_h$ vs. lag (h).

III PACF plot: Partial correlation of ε_t and ε_{t-h}, controlling for the intermediate errors, i.e., $\varepsilon_{t-1}, \ldots, \varepsilon_{t-h+1}$.

$\varepsilon_t, \varepsilon_{t-2}$ controlling ε_{t-1}. | $\cdot \underset{\sim}{X}_1, \underset{\sim}{X}_2$ controlling for $\underset{\sim}{X}_3$

Calculation of partial correlation:

$$R = ((r_{ij})) = \text{correlation matrix.}$$
$$P = R^{-1} = ((d_{ij}))$$

Partial correlation between i-th and j-th variable is $\dfrac{d_{ij}}{\sqrt{d_{ii} d_{jj}}}$.

In AR(1) model:

$$R_{3\times 3} = \sigma_\varepsilon^2 \begin{pmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{pmatrix} \qquad \sigma_\varepsilon^2 = \frac{\sigma^2}{1-\rho^2}$$

$$P = \frac{1}{\sigma_\varepsilon^2 (1-\rho^2)} \begin{bmatrix} 1 & -\rho & 0 \\ -\rho & 1+\rho^2 & -\rho \\ 0 & -\rho & 1 \end{bmatrix} = ((d_{ij}))$$

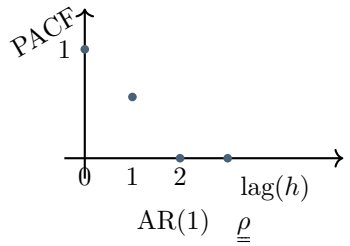
PACF of $\varepsilon_t, \varepsilon_{t-2}$ controlling for ε_{t-1} : $\dfrac{d_{13}}{\sqrt{d_{11}d_{33}}} = 0.$

$$R_T = \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{T-1} \\ \rho & 1 & \rho & \cdots & \rho^{T-2} \\ \vdots & & & & \vdots \\ \rho^{T-1} & \rho^{T-2} & \rho^{T-3} & \cdots & 1 \end{bmatrix} \qquad P = R^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & 0 & \cdots & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \cdots & 0 \\ 0 & 0 & -\rho & 1+\rho^2 & \cdots & 0 \\ \vdots & \vdots & & & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

PACF of ε_t and ε_{t-k}, controlling for $\varepsilon_{t-1}, \ldots, \varepsilon_{t-k+1}$ is 0.

$\underset{\sim}{Y} \longrightarrow \Sigma^{-1/2} \underset{\sim}{Y}$

$X \longrightarrow \Sigma^{-1/2} X$



$r_{12} \quad r_{13} \quad \ldots$

$$\underline{\underline{r_{1k}}} = \frac{\sum_{t=k+1}^{T} e_t e_{t-k}}{\sum_{t=1}^{T} e_t^2}$$

$$R = \begin{bmatrix} 1 & r_{12} & r_{13} & \cdots \\ r_{12} & 1 & r_{12} & \cdots \\ \vdots & \vdots & \vdots & \end{bmatrix}$$

AR(2): $\varepsilon_t = \phi_1 \varepsilon_{t-1} + \phi_2 \varepsilon_{t-2} + u_t$

2.3 Durbin-Watson Test

Test: Duribin-Watson Test

Under the AR(1) framework, it tests if $H_0 : \rho = 0$ or $H_1 : \rho \neq 0$.

$$\begin{aligned} D &= \frac{\sum_{t=2}^{T} (e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2} \\ &= \frac{\sum_{t=2}^{T} e_t^2 + \sum_{t=2}^{T} e_{t-1}^2 - 2\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=1}^{T} e_t^2} \quad \in [0,4] \\ &\approx 2(1-r_1) \qquad\qquad r_1 = \frac{\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=1}^{T} e_t^2} \end{aligned}$$

Under H_0, the exact distrbn. of D depends on the design matrix.

$$R = \frac{D}{4} \text{ approximately follows } \underline{\text{Beta}\,(k_0, k_1)}$$

$$k_1 + k_0 = \frac{E(D)\left[4 - E(D)\right]}{\text{Var}(D)}$$

$$k_0 = \frac{1}{4}\left(k_0 + k_1\right) E(D)$$

$$\underline{r_1 = \frac{\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=1}^{n} e_t^2}} \approx (1 - D/2) \quad \xrightarrow{H_0} \quad N(0,1) \quad \text{for } T \text{ is large.} \qquad [\text{Chatfield}]$$

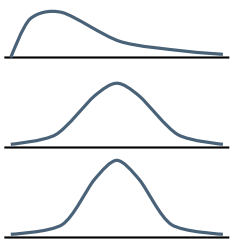
3 Normality Assumption & Departure from Normality

Normality Assumption:

$$\underset{\sim}{\varepsilon} \sim N\left(\underset{\sim}{0}, \sigma^2 I\right) \quad \times$$

$$\underset{\sim}{\varepsilon} \sim F\left(\underset{\sim}{0}, \sigma^2 I\right)$$

If the errors are heavy tailed then efficiency of OLS estimators are affected.



$X \sim \text{Cauchy}(0,1)$

$E(X)$ does not exists.

$E(|X|) = \infty$

$E\left(\underline{|X|^{\lambda}}\right)$ exists, $\quad \lambda \in (0,1)$.

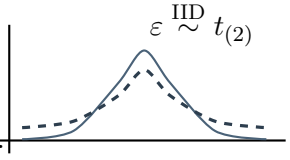


Figure 3: Comparison of light-tailed, skewed, and heavy-tailed error distributions (Cauchy$(0,1)$ and $t_{(2)}$).

Box-Cox:

$$Y^{(\lambda)} = g(Y, \lambda) = \underset{\sim}{x}^T \underset{\sim}{\beta} + \beta_0 + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2).$$

$$g(Y,\lambda) = \begin{cases} \dfrac{Y^{\lambda} - 1}{\lambda} & \text{for } \lambda \neq 0 \\ \log Y & \text{otherwise} \end{cases} \qquad \underline{\underline{\lambda \in [0,2]}}$$

3.1 Quantile-Quantile (Q-Q) Plot

Plot Quantile-quantile plot.

$$\underset{\sim}{e} \sim N\left(\underset{\sim}{0}, \sigma^2 \underline{(I-P)}\right)$$

Compare $u_i = \dfrac{e_i}{\hat{\sigma}_{UE}}$; $i = 1, 2, \ldots, n$ with $N(0,1)$.

- Order u_i's as $u_{(1)} \leq u_{(2)} \leq \cdots \leq u_{(n)}$.
- Compare $u_{(k)}$ with the $\dfrac{k-0.5}{n}$-th quantile of $N(0,1)$.

$X_i \overset{\text{IID}}{\sim} F$: known.

$X_{(1)}, X_{(2)}, \ldots, X_{(n)}$

$X_{(k)}$: $\dfrac{k}{n}$-th sample quantile.

Compare $X_{(k)}$ with $\dfrac{k}{n}$-th quantile of F.

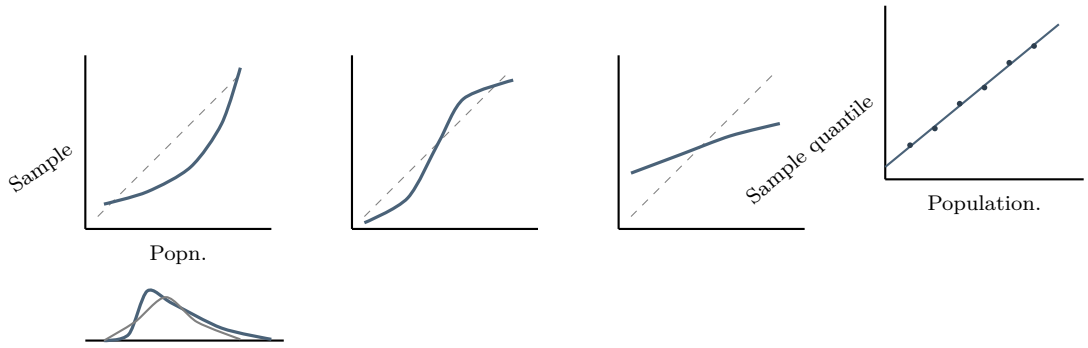


Figure 4: Q-Q plot patterns comparing sample quantiles against population quantiles.

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3.2 Detailed Graphical Analysis of Departure from Normality

Departure from normality —

Graphical tool: Q-Q plot.

Distn. of $\underset{\sim}{e}/\sigma$ is approximately $N(\underset{\sim}{0}, I)$.

$\frac{e_i}{\sigma}$'s are approximately iid $N(0,1)$.

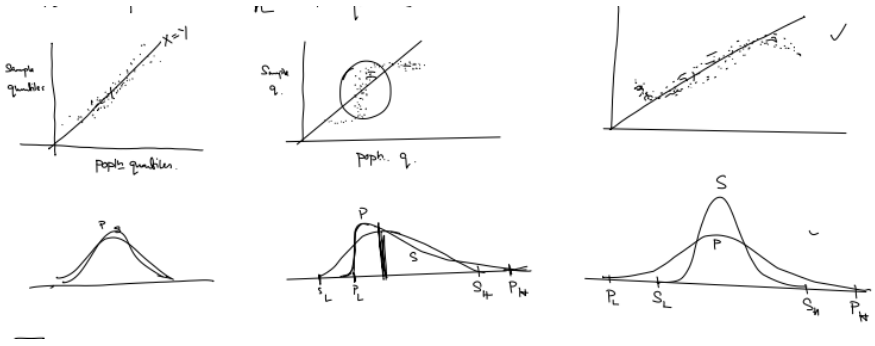
Typically, $u_i = \frac{e_i}{\hat{\sigma}_{UE}}$'s are used for a Q-Q plot.

Comparing the ordered residuals $u_{(1)} \leq \cdots \leq u_{(n)}$ with appropriate quantiles of $N(0,1)$.

$u_{(r)}$ is compared with $\frac{r-0.5}{n}$-th quantile.

$$P(X) = ((p_{ij}))$$
$$p_{ij} \sim -\frac{1}{n}$$
$$p_{ii} \sim \frac{p}{n}$$
$$p \ll n.$$

$$\underset{\sim}{Z} \sim N\left(\underset{\sim}{0}, \Sigma\right)$$



3.3 Transformations for Normality

Transformation

$$g(Y, \lambda) = Y^{(\lambda)} \sim N\left(X\underset{\sim}{\beta}, \sigma^2 I\right)$$

I) Box-Cox:

$$Y^{(\lambda)} = \begin{cases} \dfrac{Y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log Y & \text{if } \lambda = 0 \end{cases} \qquad \lambda \text{ usually varies } [-2, 2]$$

Note:

① It can be applied for non-neg. Y only.

② Except $\lambda = 0$, $Y^{(\lambda)}$ does not range in whole $\mathbb{R}$, so the normality can't hold exactly.

II) John-Draper:

$$g(Y, \lambda) = Y^{(\lambda)} = \begin{cases} \dfrac{\text{sign}(Y)}{\lambda}\left\{(1 + |Y|)^\lambda - 1\right\} & \text{for } \lambda \neq 0 \\ \text{sign}(Y)\log(1 + |Y|) \end{cases}$$

Note: This transformatn. works well when the error distn. is heavy tailed.

III) Carrol & Ruppert:

Recommends transforming both Y and $\underset{\sim}{x}^T\underset{\sim}{\beta}$ with the same function.

$$g(Y_i,\lambda) \sim N\Big(\underbrace{g\left(\underset{\sim i}{x}^T\underset{\sim}{\beta},\lambda\right)}, \ \sigma^2 I\Big)$$

$$E(Y_i) = \underset{\sim i}{x}^T\underset{\sim}{\beta}$$

$$E[g(Y_i)] = g\left(\underset{\sim i}{x}^T\underset{\sim}{\beta}\right).$$

The g can be a Box-Cox or John-Draper's transformatn.

3.4 How to Get λ? (Profile Likelihood Approach)

How to get λ?

Consider the revised model $\underline{\underline{Y_i^{(\lambda)}}} \overset{\text{IND}}{\sim} N\left(\underset{\sim i}{x}^T\underset{\sim}{\beta}, \sigma^2\right)$.

What is the distn. of Y?

Consider Box-Cox:

$$f_{Y_i}(y_i) = f_{Y_i^{(\lambda)}}\left(y_i^{(\lambda)}\right)\cdot y_i^{\lambda-1}$$

$$f_{\underset{\sim}{Y}}(\underset{\sim}{y}) = \prod_{i=1}^{n} f_{Y_i}(y_i).$$

$$l_n\left(\underset{\sim}{\beta},\sigma^2,\lambda\right) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\Big(y_i^{(\lambda)} - \underbrace{\underset{\sim i}{x}^T\underset{\sim}{\beta}}_{\uparrow\ g(\underset{\sim i}{x}^T\underset{\sim}{\beta},\lambda)}\Big)^2$$

$$+ (\lambda-1)\sum_{i=1}^{n}\log y_i$$

$$z_i = \frac{Y_i^{\lambda}-1}{\lambda} \qquad \lambda \neq 0$$

$$\frac{dz}{dy} = \frac{1}{\lambda}\left\{\lambda y_i^{\lambda-1}\right\}$$

$$= y_i^{\lambda-1}$$

$$z_i = \log Y_i \qquad \lambda = 0$$

$$\frac{dz}{dy} = \frac{1}{y_i}$$

Profile Likelihood Procedure for λ

Profile likelihood:

① First maximize the likelihood w.r.t. σ^2, keeping $\left(\underset{\sim}{\beta}, \lambda\right)$ fixed.

$$\frac{\partial l_n}{\partial \sigma^2} = 0 \implies -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}\Big(y_i^{(\lambda)} - \underbrace{\underset{\sim i}{x^T}\underset{\sim}{\beta}}_{g(\underset{\sim i}{x^T}\underset{\sim}{\beta},\lambda)}\Big)^2 = 0$$

$$\implies \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\Big(y_i^{(\lambda)} - \underbrace{\underset{\sim i}{x^T}\underset{\sim}{\beta}}_{g(\underset{\sim i}{x^T}\underset{\sim}{\beta},\lambda)}\Big)^2 = \hat{\sigma}^2_{ML}\left(\underset{\sim}{\beta}, \lambda\right).$$

The profile likelihood:

$$l_{p,n}\left(\underset{\sim}{\beta}, \lambda\right) = \max_{\sigma^2} l_n\left(\underset{\sim}{\beta}, \sigma^2, \lambda\right) = \underbrace{-\frac{n}{2}\log(2\pi) - \frac{n}{2}}_{C} - \frac{n}{2}\log\left(\hat{\sigma}^2_{ML}\left(\underset{\sim}{\beta}, \lambda\right)\right) + (\lambda - 1)\sum_{i=1}^{n}\log y_i$$

② Fix λ in a grid, usually in $[-2, 2]$, say $\lambda = \lambda_0$.
Then maximize $l_{p,n}\left(\underset{\sim}{\beta}, \lambda_0\right)$ w.r.t. $\underset{\sim}{\beta}$. Suppose $l_{p,n}\left(\underset{\sim}{\beta}, \lambda_0\right)$ is maximized at $\hat{\underset{\sim}{\beta}}(\lambda_0) =$?? (numerical optimisatn. if Carroll & Ruppert). Then compute the maximized likelihood $\underbrace{l_{p,n}\left(\hat{\underset{\sim}{\beta}}(\lambda_0), \lambda_0\right)}_{= l^*_{p,n}(\lambda_0)}$.

③ Compare $l_{p,n}\left(\hat{\underset{\sim}{\beta}}(\lambda_0), \lambda_0\right)$ for different values of λ_0, and choose that λ_0, say λ^*, which maximizes $l^*_{p,n}(\lambda_0)$.

4 Multicollinearity and Variable Selection

Problem: Correlated covariates.

Usual soln.s:

1) Reduce dimension by dropping some covariates. $\overset{n\times(p+1)}{X}$

2) Reduce dimension without dropping any particular predictor. $p + 1 > n$
(PCA, Ridge).

4.1 Variable Selection Approaches

1) Variable Selection:

Suppose we have k-many predictors, k is large: $\{x_1, \ldots, x_k\}$.

1) Screening.

2) All possible regressions (APR):

Here one consider all possible subsets of $\{1, \ldots, k\}$ and forms a model.

Model:

(I) $\underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \underset{\sim}{\varepsilon}$.

(II) $\left.\begin{array}{l} \underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim_1}{x} + \underset{\sim}{\varepsilon} \\ \underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_2 \underset{\sim_2}{x} + \underset{\sim}{\varepsilon} \\ \vdots \end{array}\right\}$ all models with 1 predictor included (k-models)

(III) $\left.\begin{array}{l} \underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim_1}{x} + \beta_2 \underset{\sim_2}{x} + \underset{\sim}{\varepsilon} \\ \underset{\sim}{Y} = \beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim_1}{x} + \beta_3 \underset{\sim_3}{x} + \underset{\sim}{\varepsilon} \\ \vdots \end{array}\right\}$ all possible models with 2 predictors included (${}^kC_2$ models)

Proceeding similarly, we have 2^k models in total.

Next, we compare these 2^k models w.r.t. some goodness of fit criterion (GFC).

Wednesday, November 4, 2026

5 Model Selection

$$\begin{aligned} \underset{\sim}{Y} &\sim N\left(\underset{\sim}{\mu}, \sigma^2 I\right) \\ \underset{\sim}{\mu} &= g(X) \\ \text{Assume} \quad g(X) &= X\underset{\sim}{\beta}. \end{aligned}$$

We have k-many predictors, and $\overset{n\times(k+1)}{X} = \left[\underset{\sim}{1} \quad \underset{\sim_1}{x} \quad \cdots \quad \underset{\sim_k}{x}\right]$, k is large.

Q: What is the best subset of the predictors?

- **APR:**

$$\mathcal{M} = \left\{ \underbrace{\left\{\underset{\sim}{1}\right\}}_{\text{Null model}}, \left\{\underset{\sim}{1}, \underset{\sim_1}{x}\right\}, \left\{\underset{\sim}{1}, \underset{\sim_2}{x}\right\}, \ldots, \underbrace{\left\{\underset{\sim}{1}, \underset{\sim_1}{x}, \ldots, \underset{\sim_k}{x}\right\}}_{\text{Full model}} \right\}, \qquad |\mathcal{M}| = 2^k$$

- **Screening:**

$$\begin{aligned} \mathcal{M}_S &= \text{all nested models} \\ &= \left\{ \underline{\left\{\underset{\sim}{1}\right\}, \left\{\underset{\sim}{1}, \underset{\sim_1}{x}\right\}, \left\{\underset{\sim}{1}, \underset{\sim_1}{x}, \underset{\sim_2}{x}\right\}}, \ \ldots \right\} \\ &\qquad \left\{\underset{\sim}{1}, \underset{\sim_2}{x}\right\}, \left\{\underset{\sim}{1}, \underset{\sim_2}{x}, \underset{\sim_1}{x}\right\} \\ &\qquad \left\{\underset{\sim}{1}, \underset{\sim_k}{x}\right\} \end{aligned}$$

$$M = \left\{ \underset{\sim}{1}, \underset{\sim}{x}_1 \right\}$$
$$\downarrow$$
$$\underset{\sim}{Y} \sim N\left(\underbrace{\beta_0 \underset{\sim}{1} + \beta_1 \underset{\sim}{x}_1}, \ \sigma^2 I \right)$$

Q. What are the criteria using which we compare the models?

Fitting & Complexity.

Bias $\propto$ Variance.

5.1 Prediction Error (PE) and Expected Model Error

Prediction Error (PE):

- $\underline{\left(X, \underset{\sim}{Y}\right)}$: Data (Training data)
- $\binom{m \times k}{X_0}, \underset{\sim 0}{\overset{m \times 1}{Y}}$: future observatn. (Testing data) $\quad \underset{\sim 0}{Y} \sim N\Big(\underset{\sim 0}{\mu} \ , \ \sigma^2 I \Big)$

$\underset{\sim 0}{\mu} \parallel$ Not true $\left(\cancel{\underset{\sim 0}{\mu} = X_0 \underset{\sim}{\beta}} \right)$?

Use the training data to estimate $\underset{\sim M}{\beta}$; $M \in \mathcal{M}$, say, $\underset{\sim OLS,M}{\hat{\beta}}$.

$\left(\text{Under model } M, \ \underset{\sim 0}{Y} \text{ is estimated as } X_{0,M} \underset{\sim OLS,M}{\hat{\beta}} \right)$

Predict $\underset{\sim 0}{Y}$ based on that:

$$PE_M = \left\| \underset{\sim 0}{Y} - X_{0,M} \underset{\sim OLS,M}{\hat{\beta}} \right\|^2 = \left\| \underset{\sim 0}{Y} - \underset{\sim 0}{\mu} + \underset{\sim 0}{\mu} - X_{0,M} \underset{\sim OLS,M}{\hat{\beta}} \right\|^2$$

$$E_{\underset{\sim 0}{Y}}(PE_M) = \underbrace{E_{\underset{\sim 0}{Y}} \left\| \underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right\|^2}_{\substack{\text{not related} \\ \text{to } M}} + \underbrace{\left\| \underset{\sim 0}{\mu} - X_{0,M} \underset{\sim OLS,M}{\hat{\beta}} \right\|^2}_{\text{Model error } (ME_M)} + 2 \underbrace{E_{\underset{\sim 0}{Y}} \left[\left(\underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right)^T \left(\underset{\sim 0}{\mu} - X_{0,M} \underset{\sim OLS,M}{\hat{\beta}} \right) \right]}_{\underset{\sim}{0}}$$

$$= E_{\underset{\sim 0}{Y}} \left[\left(\underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right)^T \left(\underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right) \right] + ME_M$$

$$= E_{\underset{\sim 0}{Y}} \left[\text{tr} \left(\left(\underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right) \left(\underset{\sim 0}{Y} - \underset{\sim 0}{\mu} \right)^T \right) \right] + ME_M$$

$$= \text{tr}\Big(\underbrace{\text{Var}\left(\underset{\sim 0}{Y} \right)}_{\sigma^2 I} \Big) + ME_M$$

$$= \underline{\underline{m\sigma^2}} + ME_M$$

$$E_{\underset{\sim}{Y}}(ME) = E_{\underset{\sim}{Y}}\left[\left\|\underset{\sim 0}{\mu} - X_{0,M}\left(X_M^T X_M\right)^{-1} X_M^T \underset{\hookrightarrow \underset{\sim}{Y}\sim N(\underset{\sim}{\mu},\sigma^2 I)}{\underset{\sim}{Y}}\right\|^2\right] \qquad \hat{\underset{\sim}{\beta}}_{OLS,M} = \left(X_M^T X_M\right)^{-1} X_M^T \underset{\sim}{Y}.$$

Special Case: In-Sample Expected Model Error

Special case: $X_0 = X$ and $\underset{\sim 0}{Y} = \underset{\sim}{Y}$, $\implies m = n$ & $X_{0,M} = X_M$.

$$\begin{aligned} E_{\underset{\sim}{Y}}(ME) &= E_{\underset{\sim}{Y}}\left(\left\|\underset{\sim}{\mu} - P(X_M)\left\{\underset{\sim}{\mu} + \underset{\sim}{\varepsilon}\right\}\right\|^2\right) \qquad \underset{\sim}{\varepsilon} \sim N\left(\underset{\sim}{0}, \sigma^2 I\right) \\ &= E_{\underset{\sim}{Y}}\left(\left\|(I - P(X_M))\,\underset{\sim}{\mu} - P(X_M)\underset{\sim}{\varepsilon}\right\|^2\right) \\ &= \underset{\sim}{\mu}^T (I - P(X_M))\,\underset{\sim}{\mu} + \underbrace{E\left(\underset{\sim}{\varepsilon}^T P(X_M)\underset{\sim}{\varepsilon}\right)}_{\mathrm{tr}(\sigma^2 P(X_M))} \\ &= \underbrace{\underset{\sim}{\mu}^T (I - P(X_M))\,\underset{\sim}{\mu}}_{\text{fit}} + \underbrace{\sigma^2 (p_M + 1)}_{\text{Complexity}} = \sigma^2 (p_M + 1) \qquad \circledast \end{aligned}$$

The criteria we will learn will try to estimate this expected model error.

5.2 Model Selection Criteria

(I) Scaled RSS

$$RSS_M = \left\|\underset{\sim}{Y} - \hat{\underset{\sim}{Y}}_M\right\|^2 = \left\|\underset{\sim}{Y} - P(X_M)\underset{\sim}{Y}\right\|^2 = \underbrace{\underset{\sim}{Y}^T (I - P(X_M))\,\underset{\sim}{Y}}_{\text{measure of fit}}.$$

$$\hat{\sigma}^2_{UE,M} = \frac{RSS_M}{n - p_M - 1} \longleftarrow \text{complexity penalty.}$$

One chooses a model with minimum $\hat{\sigma}^2_{UE,M}$.

(II) Adjusted R^2

$$R^2_M = \mathrm{Corr}^2\left(\underset{\sim}{Y}, \hat{\underset{\sim}{Y}}_M\right) \quad \text{— measure of fit.}$$

$$\text{Adj } R^2: \quad \bar{R}^2_M = 1 - \left(1 - R^2_M\right)\frac{n}{n - p_M - 1} \longleftarrow \text{complexity penalty.}$$

One chooses the model with highest $\bar{R}^2_M$.

Recall, $R^2 = 1 - \dfrac{RSS}{RSS_{H_0}}$ for $H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$ $\underset{\sim}{Y} \sim N\left(\beta_0 \underset{\sim}{1}, \sigma^2 I\right)$

$$RSS_{H_0} = \underset{\sim}{Y}^T \left(I - \frac{1}{n} \underset{\sim}{1}\underset{\sim}{1}^T\right) \underset{\sim}{Y}$$
$$= \sum_{i=1}^{n} \left(Y_i - \bar{Y}\right)^2 = SS_Y$$

So, under model $M \in \mathcal{M}$, $R^2_M = 1 - \dfrac{RSS_M}{SS_Y}$.

$$\underline{\underline{\bar{R}^2_M}} = 1 - \frac{RSS_M}{n - p_M - 1} \cdot \underbrace{\left(\frac{n}{SS_Y}\right)}_{\text{an estimator of } \sigma^2} \cdot \qquad \frac{RSS}{\sigma^2} \sim$$

(3) Mallow's C_p

$$E\left(RSS_M\right) = E\left(\underset{\sim}{Y}^T \left(I - P(X_M)\right) \underset{\sim}{Y}\right) \qquad \underset{\sim}{Y} \sim N\left(\underset{\sim}{\mu}, \sigma^2 I\right)$$
$$= \underset{\sim}{\mu}^T \left(I - P(X_M)\right) \underset{\sim}{\mu} + \operatorname{tr}\left(\sigma^2 \left(I - P(X_M)\right)\right)$$
$$= \underset{\sim}{\mu}^T \left(I - P(X_M)\right) \underset{\sim}{\mu} + \sigma^2 \left(n - p_M - 1\right)$$
$$= E\left(ME_M\right) + \sigma^2 \left\{n - 2\left(p_M + 1\right)\right\}$$
$$\frac{E\left(ME_M\right)}{\sigma^2} = \frac{E\left(RSS_M\right)}{\sigma^2} - n + 2\left(p_M + 1\right).$$

With this idea,

$$C_{p_M} = \frac{RSS_M}{\hat{\sigma}^2_F} + 2\left(p_M + 1\right) - n$$

$$\hat{\sigma}^2_F = \frac{\underset{\sim}{Y}^T \left(I - P(X)\right) \underset{\sim}{Y}}{n - k - 1} \quad : \text{estimate of } \sigma^2 \text{ using the full model.}$$

- If a model M fits well, then $E\left(ME_M\right)/\sigma^2 \approx p_M + 1$ as $\underset{\sim}{\mu}^T \left(I - P(X_M)\right) \underset{\sim}{\mu} \approx 0$.
- If a model M is a bad fit then $E\left(ME_M\right)/\sigma^2 > p_M + 1$.

Therefore, we compare C_{p_M} with $(p+1)$.
Among the models, which have $C_{p,M}$ close to $(p_M + 1)$, we choose the one with lowest C_p.

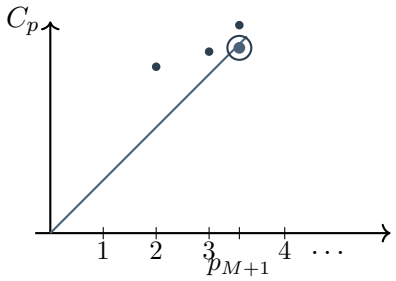


Figure 5: Mallow's C_p plot against $(p_M + 1)$, selecting the model close to the line with lowest C_p.

Relation of C_p with Adj R^2:

HW $\quad \underbrace{\dfrac{C_{p_M} - (p_M + 1)}{n - p_M - 1}} = \dfrac{1 - \bar{R}^2_M}{1 - \bar{R}^2_F} - 1 \qquad$ where $\bar{R}^2_F$: Adj R^2 based on the full model.

(4) Cross-Validation (CV)

(4) Cross-validation: If we use the same data for estimation and fitting, then overfitting takes place. $\Big($i.e., the prediction error becomes low for the training data, but remains high for a future obsn.$\Big)$.

CV comes with the idea:

① Split the data in two parts having $(n - d)$ samples and d samples, for a chosen d.

② Use the first part having $(n - d)$ samples to estimate $\underset{\sim}{\beta}$, under model M.

③ Validate model M, based on the 2nd part having d samples.

④ Consider all possible splits of $\{1, 2, \ldots, n\}$ into 2 parts having $(n - d)$ and d components, and obtain the PE_M for each such splits using ①–③.

⑤ Take an average of the PE_M's obtained in ④.

Problems:

① Choice of d. $\quad \begin{bmatrix} \text{Too small } \underline{d} \text{ is not enough to evaluate.} \\ \text{Too small } n - d \text{ may not be enough for training.} \end{bmatrix}$

② If d is large, there are $\underline{\binom{n}{d}}$ possible splits. Then it is computationally complex.
[Take N_0 many random splits.].

Leave-One-Out CV (LOOCV): $d = 1$

We use $(n - 1)$ samples for training, and $d = 1$ for testing.

Combined prediction error estimate:

$$\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \underline{\underset{\sim}{x}_{i,M}^T \hat{\underset{\sim}{\beta}}_{OLS,(-i),M}}\right)^2 = \hat{PE}_M$$

$$\left[\ \underline{\textbf{Recall:}} \quad \hat{\underset{\sim}{\beta}}_{OLS} - \hat{\underset{\sim}{\beta}}_{OLS,(-i)} = \left(X^TX\right)^{-1}\underset{\sim}{x}_i\left(\frac{e_i}{1-p_{ii}}\right) \quad ; \quad e_i = y_i - \underset{\sim}{x}_i^T\hat{\underset{\sim}{\beta}}_{OLS}\ \right]$$

$$Y_i - \underset{\sim}{x}_{i,M}^T\hat{\underset{\sim}{\beta}}_{OLS,(-i),M} = \underline{Y_i - \underset{\sim}{x}_{i,M}^T\hat{\underset{\sim}{\beta}}_{OLS,M}} + \underbrace{\underline{\underset{\sim}{x}_{i,M}^T\left(X_M^TX_M\right)^{-1}\underset{\sim}{x}_{i,M}}}_{p_{ii,M}} \cdot \frac{\left(Y_i - \underset{\sim}{x}_{i,M}^T\hat{\underset{\sim}{\beta}}_{OLS,M}\right)}{1-p_{ii,M}}.$$

6th November

LOOCV

Consider a set of $(n-1)$ samples for training and one remaining one sample for testing. We take all such subsets of $(n-1)$ sample. [n many].

Result:

The expected prediction error of obtained from LOOCV is higher than the PE when same training and testing sets are used.

PROOF:

$$\text{LOOCV-PE} = \sum_{i=1}^{n} \left(Y_i - \underset{\sim i}{x}^T \underbrace{\hat{\beta}}_{\sim \text{OLS},(-i)} \right)^2 \quad \text{— (1)}$$

$$\text{PE for same train and test sample :} \quad \sum_{i=1}^{n} \left(Y_i - \underset{\sim i}{x}^T \hat{\beta}_{\sim \text{OLS}} \right)^2$$

$$= \left\| \underset{\sim}{Y} - X \hat{\beta}_{\sim \text{OLS}} \right\|^2$$

$$= \left\| (I - P(X)) \underset{\sim}{Y} \right\|^2 \quad \text{— (2)}.$$

$$\hat{\beta}_{\sim \text{OLS},(-i)} = \hat{\beta}_{\sim \text{OLS}} - (X^T X)^{-1} \underset{\sim i}{x} \frac{e_i}{1 - p_{ii}}$$

$$e_i = Y_i - \underset{\sim i}{x}^T \hat{\beta}_{\sim \text{OLS}}$$

$$\left(Y_i - \underset{\sim i}{x}^T \hat{\beta}_{\sim \text{OLS},(-i)} \right) = e_i + p_{ii} \frac{e_i}{1 - p_{ii}} = \left(1 + \frac{p_{ii}}{1 - p_{ii}} \right) e_i = \frac{1}{1 - p_{ii}} e_i \quad \text{— (3)}$$

$$\begin{aligned}\mathbb{E}(e_i^2) &= \mathbb{E}\left[\left(Y_i - \underset{\sim}{x}_i^T \hat{\underset{\sim}{\beta}}_{\text{OLS}}\right)^2\right] \\ &= \mathbb{E}\left[(\eta_i + \xi_i)^2\right] \\ &= \eta_i^2 + \mathbb{E}(\xi_i^2) \\ &\quad + 2\eta_i \mathbb{E}(\xi_i) \\ &= \eta_i^2 + \sigma^2(1 - p_{ii})\end{aligned}$$

$$\underset{\sim}{Y} \sim N(\underset{\sim}{\mu}, \sigma^2 I)$$

where $\rightarrow$

$$\begin{aligned}\underset{\sim}{e} &= \underset{\sim}{Y} - X\hat{\underset{\sim}{\beta}}_{\text{OLS}} \\ &= \underset{\sim}{Y} - P(X)\underset{\sim}{Y} \\ &= (I - P(X))(\underset{\sim}{\mu} + \underset{\sim}{\varepsilon}) \\ &= \underbrace{(I - P(X))\underset{\sim}{\mu}}_{\underset{\sim}{\eta}} + \underbrace{(I - P(X))\overset{\underset{\sim}{\varepsilon} \sim N(0,\sigma^2 I)}{\underset{\sim}{\varepsilon}}}_{\underset{\sim}{\xi}}\end{aligned}$$

$$\mathbb{E}(\underset{\sim}{\xi}) = \underset{\sim}{0} \quad \text{as} \quad \mathbb{E}(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}$$
$$\text{Var}(\underset{\sim}{\xi}) = \sigma^2(I - P(X))$$

$$\begin{aligned}\mathbb{E}(\text{LOOCV-PE}) &= \sum_{i=1}^n \mathbb{E}\left(\frac{e_i^2}{(1-p_{ii})^2}\right) \quad \text{from } \textcircled{1} \ \& \ \textcircled{3} \\ &= \sum_{i=1}^n \frac{\eta_i^2}{(1-p_{ii})^2} + \sum_{i=1}^n \frac{\sigma^2}{(1-p_{ii})} \\ \mathbb{E}(\text{PE}) &= \mathbb{E}\left(\|(I - P(X))\underset{\sim}{Y}\|^2\right) \quad \text{from } \textcircled{2} \\ &= \sum_{i=1}^n \mathbb{E}(e_i^2) \\ &= \sum_{i=1}^n \eta_i^2 + \sum_{i=1}^n \sigma^2(1-p_{ii}).\end{aligned}$$

$$\begin{aligned}&\mathbb{E}(\text{LOOCV-PE}) - \mathbb{E}(\text{PE}) \\ &= \sum_{i=1}^n \eta_i^2 \left\{\frac{1}{(1-p_{ii})^2} - 1\right\} + \sum_{i=1}^n (1-p_{ii})\sigma^2 \left\{\frac{1}{(1-p_{ii})^2} - 1\right\}. \qquad 0 < p_{ii} < 1. \\ &> 0\end{aligned}$$

AIC

The best model should minimize the KL divergence from the true model.

Let g be the density associated with a competing model, and f be the density of the true model

then

$$\begin{aligned}\mathrm{KL}(f \parallel g) &= \int \log \frac{f(y)}{g(y)}\, \underline{f(y)\, dy} \\ &= \int \log f(y)\, f(y)\, dy - \int \log g(y)\, f(y)\, dy.\end{aligned}$$

Comparing two models w.r.t. KL-divergence is equivalent to comparing them w.r.t

$$\begin{aligned}&- \int \log g(y)\, f(y)\, dy \\ &= -\mathbb{E}_f[\log g(Y)] \\ \text{(LLN).}\quad &\approx -\frac{1}{n}\sum_{i=1}^{n} \log g(Y_i) \quad \text{— (*)}\end{aligned}$$

When a random sample of size n, $Y_1, \ldots, Y_n$ is available.

Problem arises because $g = g(\cdot, \underset{\sim}{\theta})$, and we usually substitute $\vee\underset{\sim}{\theta}$ by $\vee\underset{\sim\mathrm{ML}}{\hat{\theta}}$ using the same sample, and approximate $(*)$ by $\frac{1}{n}\sum_{i=1}^{n} \log g(Y_i, \underset{\sim\mathrm{ML}}{\hat{\theta}})$, and this leads to overfitting.

Akaike observed :

Suppose $g(\cdot, \overset{=f}{\underset{\sim 0}{\theta}})$ is true model.
In that case,

$$\mathbb{E}\left(\underline{\log g\left(\underset{\sim}{Y}, \underset{\sim\mathrm{ML}}{\hat{\theta}}(\underset{\sim}{Y})\right)}\right) - \mathbb{E}\Big(\log g(\underset{\sim}{Y}, \underset{\sim 0}{\theta})\Big) \approx \frac{p}{2}$$

$$\underset{\sim}{Y} \sim N(\underset{\sim}{\mu}, \sigma^2 I)$$

Under M: $\underset{\sim}{Y} \sim N(X_M \underset{\sim M}{\beta}, \sigma^2 I)$
where $\underset{\sim}{\theta}$ has p components.
(asymptotic result.)

Also,

$$\underbrace{\mathbb{E}\Big(\log g(\underset{\sim 0}{Y}, \underset{\sim\mathrm{ML}}{\hat{\theta}}(\underset{\sim}{Y}))\Big)}_{\Delta} - \mathbb{E}\Big(\log g(\underset{\sim}{Y}, \underset{\sim 0}{\theta})\Big) \approx -\frac{p}{2}.$$

Where $\underset{\sim 0}{Y} \perp\!\!\!\perp \underset{\sim}{Y}$.
$\underset{\sim 0}{Y}$ has same dist. as $\underset{\sim}{Y}$.
$\underset{\sim 0}{Y}$ is a test sample.

So,

$$\underbrace{\mathbb{E}\Big(\log g(\underset{\sim}{Y}, \underset{\sim\mathrm{ML}}{\hat{\theta}}(\underset{\sim}{Y}))\Big)}_{\text{What we have}} - \underbrace{\mathbb{E}\Big(\log g(\underset{\sim 0}{Y}, \underset{\sim\mathrm{ML}}{\hat{\theta}}(\underset{\sim}{Y}))\Big)}_{\text{what we want}} \approx p.$$

This motivates taking the penalty term as p

In our case :

$$\left.\begin{aligned}&\underset{\sim}{Y} \sim N(\underset{\sim}{\mu}, \sigma^2 I) \longleftarrow \text{true distn.} \\ &\text{Under model } M: \quad \underset{\sim}{Y} \sim N(X_M \underset{\sim M}{\beta}, \sigma^2 I)\end{aligned}\right\} \text{assume } \sigma^2 \text{ to be known.}$$

$$\mathbb{E}\Big(\log g_M(\underset{\sim}{Y}, \hat{\underset{\sim}{\beta}}_{M,\mathrm{ML}})\Big) - \mathbb{E}\Big(\log g_M(\underset{\sim}{Y}_0, \hat{\underset{\sim}{\beta}}_{M,\mathrm{ML}})\Big)$$
$$= -\frac{1}{2\sigma^2}\mathbb{E}\Big\|\underset{\sim}{Y} - \underbrace{X_M \hat{\underset{\sim}{\beta}}_{\mathrm{ML}}}_{P(X_M)\underset{\sim}{Y}}\Big\|^2 + \frac{1}{2\sigma^2}\mathbb{E}\Big\|\underset{\sim}{Y}_0 - X_{M,0}\hat{\underset{\sim}{\beta}}_{\mathrm{ML}}\Big\|^2.$$

$\underset{\sim}{Y}_0 \perp\!\!\!\perp \underset{\sim}{Y}$
$\underset{\sim}{Y}_0$ has the same dist. as $\underset{\sim}{Y}$.

$$= -\frac{1}{2\sigma^2}\mathbb{E}\left[\Big\|\underline{(I - P(X_M))}(\underset{\sim}{\mu} + \underset{\sim}{\varepsilon})\Big\|^2\right] + \frac{1}{2\sigma^2}\mathbb{E}\left[\Big\|\Big(\underset{\sim}{Y}_0 \underbrace{\pm\underset{\sim}{\mu}}_{\uparrow} - \underbrace{X_{M,0}(X_M^T X_M)^{-1} X_M^T}_{W_M}\underset{\sim}{Y}\Big)\Big\|^2\right]$$

$$= -\frac{1}{2\sigma^2}\Big[\underset{\sim}{\mu}^T(I - P(X))\underset{\sim}{\mu} + \sigma^2(n - p_M - 1)\Big]$$
$$+ \frac{1}{2\sigma^2}\left[\mathbb{E}\Big[\underbrace{\|\underset{\sim}{Y}_0 - \underset{\sim}{\mu}\|^2}_{\sigma^2 n.} + \mathbb{E}\Big\|\underset{\sim}{\mu} - W_M \underbrace{\underset{\sim}{Y}}_{\underset{\sim}{\mu}+\underset{\sim}{\varepsilon}}\Big\|^2 + \underbrace{\mathbb{E}\Big[(\underset{\sim}{Y}_0 - \underset{\sim}{\mu})^T(\underset{\sim}{\mu} - W_M\underset{\sim}{Y})\Big]}_{\underbrace{\mathbb{E}(\underset{\sim}{Y}_0 - \underset{\sim}{\mu})^T}_{\underset{\sim}{0}} \mathbb{E}(\quad)}\Big]\right]$$
$$\text{as } \underset{\sim}{Y}_0 \perp\!\!\!\perp \underset{\sim}{Y}.$$

$$= -\frac{1}{2\sigma^2}\Bigg\{\underset{\sim}{\mu}^T(I - P(X))\underset{\sim}{\mu} + \sigma^2(n - p_M - 1)$$
$$- \sigma^2 n - \underbrace{\mathbb{E}\Big[\|(I - W_M)\underset{\sim}{\mu} - W_M\underset{\sim}{\varepsilon}\|^2\Big]}_{\underset{\sim}{\mu}^T(I-W_M)^T(I-W_M)\underset{\sim}{\mu} + \sigma^2\mathrm{tr}(W_M^T W_M) + \underbrace{\text{prod. term.}}_{0 \text{ as } \mathbb{E}(\underset{\sim}{\varepsilon})=\underset{\sim}{0}}}\Bigg\}$$

If $X_0 \approx X$, then $W_M \approx P(X_M)$.
In that case, the last term simplifies to $\underset{\sim}{\mu}^T(I - P(X_M))\underset{\sim}{\mu} + \sigma^2(p_M + 1)$.

$$\underset{[X_0 \approx X]}{\longrightarrow\approx} (p_M + 1).$$

AIC.

$$\underbrace{-2\log g_M\Big(\underset{\sim}{Y}, (\hat{\underset{\sim}{\beta}}_{M,\mathrm{ML}}, \hat{\sigma}^2_{\mathrm{ML}})\Big)}_{n\log(2\pi) + n\log\frac{\mathrm{RSS}_M}{n} + n} + 2(p_M + 2)$$
$$= n\log \mathrm{RSS}_M + \text{const.}$$

AIC is equivalently

$$n \log \mathrm{RSS}_M + 2(p_M + 2).$$

BIC

Schwartz : imposes a prior on $\underset{\sim}{\theta}$.

Under model M : $\underset{\sim}{Y} \sim N(X_M \underset{\sim M}{\beta}, \underline{\sigma^2 I})$

$$\underset{\sim}{\theta} = \underset{\sim M}{\beta} \qquad \text{Prior :} \quad \underset{\sim M}{\beta} \sim N(\underset{\sim}{0}, \sigma^2 I), \quad \pi(\sigma^2) \propto \frac{1}{\sigma^2}.$$
$$\text{Model prior :} \quad P(M) = \pi_M.$$

Schwartz's idea is to compute the posterior probabilities of each model and consider the one with highest posterior prob. as the best model.

$$\text{Posterior of model } M \quad : \quad P(M \mid \underset{\sim}{Y}) \propto P(M) \times \underbrace{\text{likelihood.}}_{\text{integrated likelihood.}} \iint\limits_{\substack{\sigma^2,\ \beta \\ \sim M}} g_M\Big(\underset{\sim}{Y}, (\underset{\sim M}{\beta}, \sigma^2)\Big) \pi(\underset{\sim M}{\beta} \mid \sigma^2) \pi(\sigma^2)\, d\underset{\sim M}{\beta}\, d\sigma^2$$

In this case :

$$\iint\limits_{\substack{\sigma^2,\ \beta \\ \sim M}} \frac{1}{\sigma^2} \underline{\frac{1}{(\sigma^2)^{(p_M+1)/2}}} \frac{1}{(\sigma^2)^{n/2}} \exp\left\{ -\frac{1}{2\sigma^2} \Big[\underbrace{\|\underset{\sim}{Y} - X_M \underset{\sim M}{\beta}\|^2 + \|\underset{\sim M}{\beta}\|^2}_{Q} \Big] \right\} d\underset{\sim M}{\beta}\, d\sigma^2$$

$$Q = \underset{\sim M}{\beta}^T (I + X_M^T X_M) \underset{\sim M}{\beta} - 2 \underbrace{\underset{\sim}{Y}^T X_M \underset{\sim M}{\beta}}_{\underset{\sim}{Y}^T X_M (X_M^T X_M)^- (X_M^T X_M) \underset{\sim M}{\beta}} + \|\underset{\sim}{Y}\|^2 \pm \underset{\sim}{Y}^T P(X_M) \underset{\sim}{Y}$$
$$= \left(\underset{\sim M}{\beta} - \underset{\sim \mathrm{ML}}{\hat{\beta}}\right)^T (I + X_M^T X_M) \left(\underset{\sim M}{\beta} - \underset{\sim \mathrm{ML}}{\hat{\beta}}\right) + \underset{\sim}{Y}^T (I - P(X_M)) \underset{\sim}{Y}.$$

$$\longrightarrow = \int_{\sigma^2} \frac{|I + X_M^T X_M|^{-1/2}}{(\sigma^2)^{\frac{n}{2}+1}} \exp\left\{ -\frac{1}{2\sigma^2} \underline{\underset{\sim}{Y}^T (I - P(X_M)) \underset{\sim}{Y}} \right\} d\sigma^2.$$

$$= |I + X_M^T X_M|^{-1/2} \underline{\underline{\Gamma\left(\frac{n}{2}\right)}} \left| \underbrace{\underset{\sim}{Y}^T (I - P(X_M)) \underset{\sim}{Y}}_{\mathrm{RSS}_M} \right|^{-n/2}.$$

$$\int (\sigma^2)^{-\frac{n}{2}-1} e^{-\beta/\sigma^2} d\sigma^2 = \Gamma\left(\frac{n}{2}\right) \beta^{+n/2}$$

Observe that $X_M^T X_M \approx n\Sigma_M$.

[Suppose X_j's are random, and $X_{1j}, \ldots, X_{nj}$ are iid samples from X_j ; $j = 1, \ldots, n$.

$$\mathbb{E}(\underset{\sim}{X}) = \mathbb{E}\left[\begin{pmatrix} X_1 \\ \vdots \\ X_{p_M+1} \end{pmatrix}\right] = \underset{\sim}{0} \quad \text{and} \quad \mathrm{Var}(\underset{\sim}{X}) = \Sigma.$$

$$X_M^T X_M = \left(\left(\sum_{i=1}^{n} X_{ik} X_{ij}\right)\right)$$

Then

$$\begin{aligned} -\frac{1}{2}\log|I + X_M^T X_M| = \log|I + X_M^T X_M|^{-1/2} &= -\frac{1}{2}\sum_{k=1}^{p_M+1}\log(1+n\sigma_{k,M}) \\ &\qquad \text{where } \sigma_{k,M} \text{ is the } k\text{-th} \\ &\approx -\frac{1}{2}(p_M+1)\log(1+\sigma n) \quad \text{eigenval. of } \Sigma_M. \\ &\approx -\frac{1}{2}(p_M+1)\log n. \end{aligned}$$

$$P(M \mid \underset{\sim}{Y}) \propto |I + X_M^T X_M|^{-1/2} \left(\mathrm{RSS}_M\right)^{+n/2}.$$

$$\log P(M \mid \underset{\sim}{Y}) \overset{?}{=} -\frac{1}{2}\log|I + X_M^T X_M| - \frac{n}{2}\log \mathrm{RSS}_M + \text{const.}$$

$$\underline{\overset{?}{\approx} +\frac{n}{2}\log \mathrm{RSS}_M - \frac{(p_M+1)}{2}\underline{\underline{\log n}}.}$$